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Technical Document

Trade Entry and Maintenance Web Service

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Version	Date	Reason	Who
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Introduction

The purpose of this document is to provide customers with a technical overview of the Trade Entry and Maintenance Web Service.

The Trade Entry and Maintenance Web Service allows you to send in new trades and request trade actions over HTTP and HTTPS as JSON, following the format of an Inbound Message (detailed in this document) to the server. The Server will process and serve a response to the new trades and trade action requests passed in as JSON via an Outbound Message (detailed in this document). Examples of how to authenticate and send in a new trade and trade action requests; and methods for how you wish to handle the response from the Server are also detailed in this document.

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Authentication

Authentication request should be sent to the following URL: `http://<hostname>:<port>/index.html`

where `hostname` and `port` are the location of the running `glassd` server

You must send in the following two parameters: `login_user_name` and `login_password`.

Before sending in a new trade or trade action request via the web service you must authenticate against the TA server to get a session to use for subsequent HTTP connections to the server. To do this you must open a HTTP Request to the server and send a username and password; the cookies returned by the server (passed back as a HTTP Header 'Set-Cookie') are those you will need to set on your HTTP requests (by passing them into the HTTP Header 'Cookie' on your requests).

Form Based Authentication

Form Based Authentication requires the following in the `glass.xml` configuration file (in the authentication section):

```
<allowed-mechanisms>
        <fba providers='kerberos,internal' />
</allowed-mechanisms>
```

Example request HTTP query in NodeJS

```
function authenticate()
{
        var postData = querystring.stringify({
                login_user_name: "username",
                login_password: "password"
        });

        // An object of options to indicate where to post to
        var postOptions = {
                host: 'localhost',
                port: '10080',
                path: '/index.html',
                method: 'POST',
                headers: {
                        'Content-Type': 'application/x-www-form-urlencoded',
                        'Content-Length': Buffer.byteLength(postData)
                }
        };

        var postReq = http.request(postOptions, function(res)
        {
                sendRequest(res.headers['set-cookie']);
        });

        postReq.write(postData);
        postReq.end();
}
```

Example request HTTP query in Fiddler

```
Header
        Content-Type : application/x-www-form-urlencoded
Body
        &login_user_name=username&login_password=password
```


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HTTP Authentication

HTTP Authentication requires the following in the glass.xml configuration file (in the authentication section) :

```
<allowed-mechanisms>
        <http>
                <Basic providers='internal' />
                <Digest providers='windows' />
                <NTLM providers='windows' />
                <Negotiate providers='windows' />
        </http>
</allowed-mechanisms>
```

Example request HTTP query in Fiddler

```
Header
        Content-Type : application/json; charset=utf-8
        Authorization Basic <string>
```

Where <string> is "username:password" converted to base64

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Sending a Request

Inbound requests should be sent to the following URLs:

Action	URL
New Trade	`http://<hostname>:<port>/api/TRDENTRY/Trade/NewTrade`
Trade Cancellation	`http://<hostname>:<port>/api/TRDENTRY/Trade/CancelTrade`
Trade Partial Return	`http://<hostname>:<port>/api/TRDENTRY/Trade/PartialReturnTrade`
Trade Full Return	`http://<hostname>:<port>/api/TRDENTRY/Trade/FullReturnTrade`
Trade Rate Change	`http://<hostname>:<port>/api/TRDENTRY/Trade/RateChangeTrade`
Trade Fee Change	`http://<hostname>:<port>/api/TRDENTRY/Trade/FeeChangeTrade`
Trade Spread Change	`http://<hostname>:<port>/api/TRDENTRY/Trade/SpreadChangeTrade`
Trade Copy	`http://<hostname>:<port>/api/TRDENTRY/Trade/CopyTrade`
Trade Authorize	`http://<hostname>:<port>/api/TRDENTRY/Trade/AuthorizeTrade`

Once authenticated against the server - i.e. a valid session id for the configured user has been returned - this cookie should be used in the HTTP headers of an inbound message JSON object that contains the requested trade, or trade action parameters/details. By default, this Request will return you an Outbound Message containing your Trade Response. If the Request fails and has not been processed - for example, a missing mandatory field -then an error will be returned as part of Outbound Message.

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Inbound Message

An Inbound message is a JSON Object sent to the server over HTTP. Each inbound message could contain details of a single trade or multiple trades.

Message Structure

There are common fields shared by both new trade and trade action messages, which are shown below, followed by details of specific trade messages:

General Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See table below for allowable parameters.	`"header": { "params": {...} }`
items	JSON Array	Contains one or more new trades/trade actions.	`"items": [ { } ]`

Parameters

Field	Type	Description	Optional	Example
timeout	Integer	For New Trades only. Maximum time in seconds to wait for the synchronous request. If the request times-out then you get a response with the current status of the trade entry requests.	•	`"timeout": 30`
worksheet_name	String	For New Trades only. Allows users to view trades that were sent/booked by the Web Service via the GUI. If no worksheet name is provided, it defaults	•	`"worksheet_name": "WebServiceSheet"`

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Field	Type	Description	Optional	Example
		to "WebServiceSheet".		
currency	String	Associated currency code for the action.	•	`"currency": "GBP"`
timezone	String	Associated timezone code for the action.	•	`"timezone": "UTC"`
columns	String	Specifies the columns to be included in the response. If not present, the default set of columns will be returned, which is: Trade No, Source System, Source ID, External System, External ID and Status.	•	`"columns": "worksheet_name,status,status_text, trade_type_sht_code,security_name, security_sedol,qty,cpy_id, entity_sht_code,account_sht_code, coll_type_text,trade_ccy_id, price_without_margin,price_cash_ccy, div_req,tax_pct,all_in,req_margin, start_val,fee,rebate,ref_index_sht_code, trade_date,start_date, end_date,prepay_days,dvp_text,purpose, mark_trade_text,comments,external_comments, settlement_mode_sht_code,settlement_clearer, cash_settlement_mode_sht_code, cash_settlement_clearer,trade_ext_id, avail_qty,days,daily_revenue,ind_rate, cash_start_date"`
book	String	If set to 'true' (the default), then specifies that the service request should create trades on the worksheet and then attempt to book them. If set to 'false' then specifies that the service request should create trades, but not attempt to book them.	•	`"book": "false"`
auto_request_approval	String	If set to 'true' and the trade needs to be approved, then sends the trade to the Approvals screen automatically. If set to 'false', then does not send the trade to the Approvals screen.	•	`"auto_request_approval": "true"`

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Field	Type	Description	Optional	Example
allow_security_defaults		If set to 'true', then a change of security will result in certain fields being set based on the defaults for the security. If set to 'false', then no defaults will be applied.	•	`"allow_security_defaults": "true"`

New Trade Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See Parameters above for full details.	`"header": { "params": { "currency": "GBP", "timezone": "UTC", "worksheet_name": "NewTrades", "timeout": 15, } }`
items	JSON Array	Contains one or more new trades.	`"items": [ { "request_source_id": "...", "request_source": "...", "orig_id": "...", "orig_source": "...", "security_sedol": "...", "cpy_id": "...", "start_date": "...", "fee": "...", } ]`

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Trade Cancellation Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See Parameters above for full details.	`"header": { "params": { "currency": "GBP", "timezone": "UTC", } }`
items	JSON Array	Contains one or more trade action parameters.	`"items": [ { "ext_id":"59297", "ext_sys":"G1", } ]`

Partial Return of a Trade Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See Parameters above for full details.	`"header": { "params": { "currency": "GBP", "timezone": "UTC", } }`
items	JSON Array	Contains one or more trade action parameters.	`"items": [ { "id":"59297", "qty":"100000", } ]`

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Full Return of a Trade Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See Parameters above for full details.	`"header": { "params": { "currency": "GBP", "timezone": "UTC", } }`
items	JSON Array	Contains one or more trade action parameters.	`"items": [ { "id":"59283", "trade_date":"2016-08-25", "effective date":"2016-08-27" , } ]`

Trade Rate Change Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See Parameters above for full details.	`"header": { "params": { "currency": "GBP", "timezone": "UTC", } }`
items	JSON Array	Contains one or more trade action parameters.	`"items": [ { "rebate":"0.4", "rebate_adjustment":"0.7", } ]`

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Trade Fee Change Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See Parameters above for full details.	`"header": { "params": { "currency": "GBP", "timezone": "UTC", } }`
items	JSON Array	Contains one or more trade action parameters.	`"items": [ { "fee":"0.6", "fee_adjustment":"0.8", } ]`

Trade Spread Change Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See Parameters above for full details.	`"header": { "params": { "currency": "GBP", "timezone": "UTC", } }`
items	JSON Array	Contains one or more trade action parameters.	`"items": [ { "spread":"0.1", "spread_adjustment":"0.4", } ]`

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Trade Copy Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See Parameters above for full details.	`"header": { "params": { "currency": "GBP", "timezone": "UTC", } }`
items	JSON Array	Contains one or more trade action parameters.	`"items": [ { "id": "58456", "effective_date": "2017-09-20", "worksheet": "Worksheet123" } ]`

Trade Authorize Message Structure

Field	Type	Description	Example
header	JSON Object	JSON Object containing parameters. See Parameters above for full details.	`"header": { "params": { "currency": "GBP", "timezone": "UTC", } }`
items	JSON Array	Contains one or more trade action parameters.	`"items": [ { "ext_id": "16262", "ext_sys": "Sys2", } ]`

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Message Definitions

New Trade

A JSON Object detailing a new Trade.

Field	Type	Description	Optional	Example
request_source_id	String	A unique ID supplied by the web service client.	•	`"request_source_id": "UNIQUE-MESSAGE-ID-1234"`
request_source	String	Could be web service client name. Used in combination with request_source_id for a unique reference.	•	`"request_source": "WEBAPP"`
orig_id	String	The unique source trade ID.	•	`"orig_id": "ORIG-TRADE-ID-1234"`
orig_source	String	A string indicating the source of the request.	•	`"orig_source": "Bloomberg"`
entity_sht_code	String	The entity short code for the new trade.	•	`"entity_sht_code": "Entity1"`
trd_type_sht_code	String	The unique identifier to identify the trade type, e.g., "L" - Loan, "B" - Borrow.		`"trd_type_sht_code": "L"`
sec_id	String	Security for this new trade. Can be either a SEDOL, ISIN, CUSIP, Security Name etc. System will find the		`"sec_id": "SUEZF"`

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Field	Type	Description	Optional	Example
		best match against the available security fields for the string entered.		
security_sedol	String	Security SEDOL for this new trade.	•	`"security_sedol": "2848004"`
security_isin	String	Security ISIN for this new trade.	•	`"security_isin": "CA0002551095"`
security_cusip	String	Security CUSIP for this new trade.	•	`"security_cusip": "000255109"`
cpy_sht_code	String	The Counterparty short code for this new trade.		`"cpy_sht_code": "CPY1"`
account_sht_code	String	The Account short code for this new trade.	•	`"account_sht_code": "ACC1"`
qty	Integer	The quantity required for this new trade		`"qty": "10000"`
fee	Double	The fee rate for this new trade		`"fee": "0.9"`
rebate	Double	The rebate rate for this new trade		`"rebate": "0.4"`
trade_ccy_id	String	The currency code for this new trade		`"trade_ccy_id": "CAD"`
price_sec_ccy	String	The price in the Security Currency	•	`"price_sec_ccy": "16.858"`
req_margin	String	The margin required	•	`"req_margin": "105"`

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Field	Type	Description	Optional	Example
		for this new trade		
coll_type	String	The collateral type of this new trade, e.g., "C" - Cash, "N" - Non-Cash, "P" - Cash Pool	•	`"coll_type": "C"`
dvp	String	DVP for new trade	•	`"dvp": "N"`
settlement_ mode_id	String	The settlement mode required for this new trade	•	`"settlement_mode_id": "DOM"`
trade_date	String	The trade date of this new trade, yyyy-mm-dd		`"trade_date": "2016-08-24"`
start_date	String	The start date of the new trade (settlement date), yyyy-mm-dd		`"start_date": "2016-08-26"`
end_date	String	The end date of this new trade, yyyy-mm-dd	•	`"end_date": "2016-09-26"`
purpose	String	A string indicating the purpose of new trade	•	`"purpose": "Purpose1"`
comments	String	A string for any additional comments	•	`"comments": "AdditionalComment1"`
start_val	Double	A string indicating the start trade value.	•	`"start_val": "10000.45"`

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Field	Type	Description	Optional	Example
start_val	String	The cash start date of the new trade, yyyy-mm-dd.	•	`"cash_start_date": "2016-03-21"`
int_type	String	A string indicating the interest type, e.g., "B" - Both, "F" - First Day, "L" - Last Day, etc.	•	`"int_type": "B"`
profit_centre	String	A string indicating the profit centre.	•	`"profit_centre": "A"`

Cancellation of a Trade

A JSON Object detailing a trade cancellation.

Field	Type	Description	Optional	Example
id	Integer	ID of the trade.	•	`"id": "55"`
ext_sys	String	External system identifier.	•	`"ext_sys": "G1"`
ext_id	Integer	External identifier of trade.	•	`"ext_id": "55"`
src_id	String	Source Trade ID.	•	`"src_id": "UNIQUE-MESSAGE-ID-1234"`
src_sys	String	Source system.	•	`"src_sys": "TA"`
src_id_2	String	Secondary Source Trade ID.	•	`"src_id_2": "UNIQUE-MESSAGE-ID-4567"`

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Field	Type	Description	Optional	Example
src_sys_2	String	Secondary Source system.	•	"src_sys_2": "XY"

Partial Return of a Trade

A JSON Object detailing a partial trade return.

Field	Type	Description	Optional	Example
id	Integer	ID of the trade.	•	"id": "55"
ext_sys	String	External system identifier.	•	"ext_sys": "G1"
ext_id	Integer	External identifier of trade.	•	"ext_id": "55"
src_id	String	Source Trade ID.	•	"src_id": "UNIQUE-MESSAGE-ID-1234"
src_sys	String	Source system.	•	"src_sys": "TA"
src_id_2	String	Secondary Source Trade ID.	•	"src_id_2": "UNIQUE-MESSAGE-ID-4567"
src_sys_2	String	Secondary Source system.	•	"src_sys_2": "XY"
trade_date	String	Trade date.	•	"trade_date: "2016-03-21"
effective_date	String	Trade effective date.	•	"effective_date": "2017-03-22"
qty	Integer	Trade quantity (must be		"qty": "100000"

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Field	Type	Description	Optional	Example
		>0).		
do_not_instruct	String	Do not instruct flag (defaults to false).	•	`"do_not_instruct": "TRUE"`

Full Return of a Trade

A JSON Object detailing a full trade return.

Field	Type	Description	Optional	Example
id	Integer	ID of the trade.	•	`"id": "55"`
ext_sys	String	External system identifier.	•	`"ext_sys": "G1"`
ext_id	Integer	External identifier of trade.	•	`"ext_id": "55"`
src_id	String	Source Trade ID.	•	`"src_id": "UNIQUE-MESSAGE-ID-1234"`
src_sys	String	Source system.	•	`"src_sys": "TA"`
src_id_2	String	Secondary Source Trade ID.	•	`"src_id_2": "UNIQUE-MESSAGE-ID-4567"`
src_sys_2	String	Secondary Source system.	•	`"src_sys_2": "XY"`
trade_date	String	Trade date.	•	`"trade_date: "2016-03-21"`

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Field	Type	Description	Optional	Example
effective_date	String	Trade effective date.	•	`"effective_date": "2017-03-22"`
do_not_instruct	String	Do not instruct flag (defaults to false).	•	`"do_not_instruct": "TRUE"`

Rate Change of a Trade

A JSON Object detailing a trade rate change.

Field	Type	Description	Optional	Example
id	Integer	ID of the trade.	•	`"id": "55"`
ext_sys	String	External system identifier.	•	`"ext_sys": "G1"`
ext_id	Integer	External identifier of trade.	•	`"ext_id": "55"`
src_id	String	Source Trade ID.	•	`"src_id": "UNIQUE-MESSAGE-ID-1234"`
src_sys	String	Source system.	•	`"src_sys": "TA"`
src_id_2	String	Secondary Source Trade ID.	•	`"src_id_2": "UNIQUE-MESSAGE-ID-4567"`
src_sys_2	String	Secondary Source system.	•	`"src_sys_2": "XY"`
rebate	Double	Rebate rate for the trade.		`"rebate": "0.4"`
rebate_adjustment	Double	Rebate adjustment		`"rebate_adjustment": "0.7"`

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Field	Type	Description	Optional	Example
		applicable.		
effective_date	String	Rate change effective date.	•	`"effective_date": "2017-03-22"`
recalc_fda	String	Whether or not the FDA for the trade should be recalculated.	•	`"recalc_fda": "TRUE"`
rate_outlier	String	Whether or not the trade is an outlier for indicative rate calculations.	•	`"rate_outlier": "TRUE"`

Fee Change of a Trade

A JSON Object detailing a trade fee change.

Field	Type	Description	Optional	Example
id	Integer	ID of the trade.	•	`"id": "55"`
ext_sys	String	External system identifier.	•	`"ext_sys": "G1"`
ext_id	Integer	External identifier of trade.	•	`"ext_id": "55"`
src_id	String	Source Trade ID.	•	`"src_id": "UNIQUE-MESSAGE-ID-1234"`
src_sys	String	Source system.	•	`"src_sys": "TA"`

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Field	Type	Description	Optional	Example
src_id_2	String	Secondary Source Trade ID.	•	`"src_id_2": "UNIQUE-MESSAGE-ID-4567"`
src_sys_2	String	Secondary Source system.	•	`"src_sys_2": "XY"`
fee	Double	Fee for the trade.		`"fee": "0.6"`
fee_adjustment	Double	Fee adjustment applicable.		`"fee_adjustment": "0.8"`
effective_date	String	Fee change effective date.	•	`"effective_date": "2017-03-22"`
recalc_fda	String	Whether or not the FDA for the trade should be recalculated.	•	`"recalc_fda": "TRUE"`
rate_outlier	String	Whether or not the trade is an outlier for indicative rate calculations.	•	`"rate_outlier": "TRUE"`

Spread Change of a Trade

A JSON Object detailing a trade spread change.

Field	Type	Description	Optional	Example
id	Integer	ID of the trade.	•	`"id": "55"`
ext_sys	String	External system identifier.	•	`"ext_sys": "G1"`

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Field	Type	Description	Optional	Example
ext_id	Integer	External identifier of trade.	•	`"ext_id": "55"`
src_id	String	Source Trade ID.	•	`"src_id": "UNIQUE-MESSAGE-ID-1234"`
src_sys	String	Source system.	•	`"src_sys": "TA"`
src_id_2	String	Secondary Source Trade ID.	•	`"src_id_2": "UNIQUE-MESSAGE-ID-4567"`
src_sys_2	String	Secondary Source system.	•	`"src_sys_2": "XY"`
spread	Double	Spread for the trade.		`"spread": "3.0"`
spread_ adjustment	Double	Spread adjustment applicable.		`"spread_adjustment": "0.4"`
recalc_fda	String	Whether or not the FDA for the trade should be recalculated.	•	`"recalc_fda": "TRUE"`
rate_outlier	String	Whether or not the trade is an outlier for indicative rate calculations.	•	`"rate_outlier": "TRUE"`

Authorise a Trade

A JSON Object detailing a trade authorisation.

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Field	Type	Description	Optional	Example
id	Integer	ID of the trade. You specify this, *OR*ext_id and ext_sys; OR src_id and src_sys; OR src_id_2 and src_sys_2.	•	`"id": "55"`
ext_sys	String	External system identifier.	•	`"ext_sys": "G1"`
ext_id	Integer	External identifier of trade.	•	`"ext_id": "55"`
src_id	String	Source Trade ID.	•	`"src_id": "UNIQUE-MESSAGE-ID-1234"`
src_sys	String	Source system.	•	`"src_sys": "TA"`
src_id_2	String	Secondary Source Trade ID.	•	`"src_id_2": "UNIQUE-MESSAGE-ID-4567"`
src_sys_2	String	Secondary Source system.	•	`"src_sys_2": "XY"`

Copy a Trade

A JSON Object detailing a trade copy.

Field	Type	Description	Optional	Example
id	Integer	ID of the trade.	•	`"id": "55"`
ext_sys	String	External system identifier.	•	`"ext_sys": "G1"`

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Field	Type	Description	Optional	Example
ext_id	Integer	External identifier of trade.	•	`"ext_id": "55"`
src_id	String	Source Trade ID.	•	`"src_id": "UNIQUE-MESSAGE-ID-1234"`
src_sys	String	Source system.	•	`"src_sys": "TA"`
src_id_2	String	Secondary Source Trade ID.	•	`"src_id_2": "UNIQUE-MESSAGE-ID-4567"`
src_sys_2	String	Secondary Source system.	•	`"src_sys_2": "XY"`
quantity	Integer	Trade quantity (must be >0).		`"qty": "100000"`
effective_date	String	Copy trade effective date.	•	`"effective_date": "2017-03-22"`
do_not_instruct	String	Do not instruct flag (defaults to false).	•	`"do_not_instruct": "TRUE"`
worksheet	String	Worksheet to which to copy the trade.	•	`"worksheet": "Worksheet123"`
trade_date	String	Trade date.	•	`"trade_date: "2016-03-21"`

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Inbound Message Examples

New Trade

HTTP Header

```
Content-Length: 348
Content-Type: application/json
Cookie: <Cookie>
```

Body


```
{

   "header": {

         "params": {"timeout": 30,
                    "worksheet_name": "WebServiceSheet"
               }
    },


 "items": [
               {
                       "request_source_id": "UNIQUE-MESSAGE-ID-1234",
                       "request_source": "WEBAPP",
                       "orig_id": "SOURCE-TRADE-ID-1234",
                       "orig_source": "Bloomberg",
                       "entity_sht_code": "Entity1",
                       "trd_type_sht_code": "L",
                       "sec_id": "GT",
                       "cpy_sht_code": "CPY001",
                       "account_sht_code": "ACC1",
                       "qty": 10000,
                       "fee": 0.25,
                       "trade_ccy_id": "USD",
                       "price_sec_ccy": "USD",
                       "req_margin": "105",
                       "coll_type": "N",
                       "settlement_mode_id": "DOM",
                       "trade_date": "2016-08-25",
                       "start_date": "2016-08-29"
               }
       ]
)
```


HTTP Header

```
Content-Length: 348
Content-Type: application/json
Cookie: <Cookie>
```

Body


```
{
```

```
  "header": {
        "params": {"timeout": 30,
                   "worksheet_name": "NewTrades20160825"
              }
   },


"items": [
                 {
                      "request_source_id": "UNIQUE-MESSAGE-ID-1234",
                      "request_source": "WEBAPP",
                      "orig_id": "SOURCE-TRADE-ID-1234",
                      "orig_source": "Bloomberg",
                      "entity_sht_code": "Entity1",
                      "trd_type_sht_code": "L",
                      "security_cusip": "001092105",
                      "cpy_id": "CPY002",
                      "account_id": "ACC1",
                      "qty": 10000,
                      "rebate": 0.25,
                      "trade_ccy_id": "CAD",
                      "price_sec_ccy": "CAD",
                      "req_margin": 105,
                      "coll_type": "C",
                      "settlement_mode_id": "DOM",
                      "trade_date": "2016-08-25",
                      "start_date": "2016-08-29",
                      "purpose": "Purpose1",
                      "comments": "Comment1"
              },
              {

                      "request_source_id": "UNIQUE-MESSAGE-ID-1235",
                      "request_source": "WEBAPP",
                      "orig_id": "SOURCE-TRADE-ID-1235",
                      "orig_source": "Bloomberg",
                      "entity_id": "Entity1",
                      "trd_type_sht_code": "L",
                      "security_cusip": "000255109",
                      "cpy_sht_code": "CPY003",
                      "account_sht_code": "ACC1",
                      "qty": 50000,
                      "rebate": 0.25,
                      "trade_ccy_id": "CAD",
                      "price_sec_ccy": "CAD",
                      "req_margin": 105,
                      "coll_type": "C",
                      "settlement_mode_id": "DOM",
                      "trade_date": "2016-08-25",
                      "start_date": "2016-08-29",
                      "purpose": "Purpose1",
                      "comments": "Comment1"
              }
      ]
}
```


Trade Cancellation

Cancelling a Global One trade using the bargain reference (in ‘ext_id’):

HTTP Header

```
Content-Length: 348
Content-Type: application/json
Cookie: <Cookie>
```

Body

```
    {
     "header":{
           "params":{"currency":"GBP",
                     "timezone":"UTC"
                }
    },
 "items":[
           {
                        "ext_id": "59297",
                        "ext_sys": "G1"
            }
       ]
}
```

Partial Return of a Trade

Partial return of 100000 shares:

HTTP Header

```
Content-Length: 348
Content-Type: application/json
Cookie: <Cookie>
```

Body

```
    {
     "header":{
           "params":{"currency":"GBP",
                     "timezone":"UTC"
                }
    },
 "items":[
           {
                        "id": "59297",
                        "qty": "100000"
            }
       ]
}
```

Full Return of a Trade

Full return of a trade:

HTTP Header

```
Content-Length: 348
Content-Type: application/json
Cookie: <Cookie>
```


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Body

```
    {
     "header":{
           "params":{"currency": "GBP",
                     "timezone": "UTC"
                     }
    },
 "items":[
           {
                       "id": "59283",
                       "trade_date": "2016-08-25",
                       "effective date": "2016-08-27"
            }
       ]
}
```

Trade Rate Change

Changing the rate for a Global One trade using the bargain reference (in 'ext_id'):

HTTP Header

```
       Content-Length: 348
       Content-Type: application/json
       Cookie: <Cookie>
```

Body

```
    {
     "header":{
           "params":{"currency": "GBP",
                     "timezone": "UTC"
                }
    },
 "items":[
           {
                       "id": "59297",
                       "effective_date": "2017-09-20",
                       "rebate": "0.4"
            }
       ]
}
```

Trade Fee Change

Changing the fee for a Global One trade using the bargain reference (in 'ext_id'):

HTTP Header

```
       Content-Length: 348
       Content-Type: application/json
       Cookie: <Cookie>
```

Body

```
    {
     "header":{
           "params":{"currency": "GBP",
                     "timezone": "UTC"
```


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```
                }
    },
 "items":[
           {
                        "id": "58982",
                        "effective_date": "2017-09-20",
                        "fee": "80"

            }
       ]
}
```


Trade Spread Change

Changing the spread for a Global One trade using the bargain reference (in 'ext_id'):

HTTP Header

```
Content-Length: 348
Content-Type: application/json
Cookie: <Cookie>
```

Body


```
    {
     "header":{
           "params":{"currency": "GBP",
                     "timezone": "UTC"
                }
    },
 "items":[
           {
                        "id": "59297",
                        "spread": "3.0"

            }
       ]
}
```


Trade Copy

Copying a trade to a Worksheet:

HTTP Header

```
Content-Length: 348
Content-Type: application/json
Cookie: <Cookie>
```

Body


```
    {
     "header":{
           "params":{"currency": "GBP",
                     "timezone": "UTC"
                }
    },
 "items":[
           {
                        "id": "58456",
```

```
                        "effective_date": "2017-09-20",
                        "worksheet": "Worksheet123"

            }
        ]
}
```

Trade Authorize

Authorizing a trade:

HTTP Header

```
Content-Length: 348
Content-Type: application/json
Cookie: <Cookie>
```

Body

```
    {
     "header":{
           "params":{"currency": "GBP",
                     "timezone": "UTC"
                }
    },
 "items":[
            {
                        "ext_id": "16262",
                        "ext_sys": "Sys2"

             }
        ]
}
```


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Outbound Message

Trade Entry

The response JSON Object sent back from the Server:

Success

Field	Type	Description	Example
id	Integer	Identifier of trade within the worksheet.	`"id": 55`
request_source_id	String	The original source Trade ID.	`"request_source_id": "UNIQUE-MESSAGE-ID-1234"`
request_source	String	The original source of the request	`"request_source": "WEBAPP"`
status	String	The status of the trade booking, for example: - ACCEPTED - INVALID	`"status": "ACCEPTED"`
status_text	String	Textual description for the status.	`"status_text": "Price Without Margin must be greater than zero."`
trade_ext_id	String	External Id of the trade booked or empty string (if not booked). Populated with the identifier of the trade in the downstream external system (e.g. Global One bargain reference).	`"trade_ext_id": "123ABC"` `"trade_ext_id": ""`
trd_id	Integer	Identifier of the trade booked, or null (if not booked).	`"trd_id": 1234` `"trd_id": null`

Failure

Field	Type	Description	Example
errorMessage	String	Textual description for an error message.	`"errorMessage": "Unable to insert new Trades into the database."`
success	String	Request action outcome.	`"success": "false"`

Trade Actions

The response JSON Object sent back from the Server is common to all trade actions:

Success

Field	Type	Description	Example
ext_id	Integer	External identifier of trade.	`"ext_id": 55`
ext_sys	String	The external system identifier.	`"ext_sys": "G1"`

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Field	Type	Description	Example
id	Integer	Trade identifier.	`"id": 59296`
src_id	String	Original source Trade ID.	`"src_id": "UNIQUE-MESSAGE-ID-1234"`
src_sys	String	Original source system.	`"src_sys": "TA"`
status	String	The status of the trade booking.	`"status": "NEW"`
status_text	String	Textual description for the status.	`"status_text": "Amendment or cancellation of Repo trades is prohibited."`

Failure

Field	Type	Description	Example
errorMessage	String	Textual description for an error message.	`"errorMessage": "The Return Quantity value must be greater than zero."`
success	String	Request action outcome.	`"success": "false"`

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Outbound Message Examples

Successfully Booked Trade


```
{
 "items": [
          {
               "request_source_id": "UNIQUE-MESSAGE-1234",
               "request_source": "WEBAPP",
               "id": "1255",
               "trd_id": "1234",
               "trade_ext_id": "EXT-TRADE-ID-1234",
               "status": "BOOKED"
          }
 ],
       "success": "true",
       "total_count": "1",
       "version": "1"
}
```


Failed to Book Trade


```
{
 "items": [
          {
               "request_source_id": "UNIQUE-MESSAGE-1234",
               "request_source": "WEBAPP",
               "id": "null",
               "trd_id": "null",
               "trade_ext_id": "",
               "status": "INVALID",
               "status_text": "The Quantity value must be greater than zero"
          }
 ],
       "success": "true",
       "total_count": "1",
       "version": "1"
}
```


Trade preliminary booked in TA but requires approval before it can be booked and sent to external system


```
{
 "items": [
          {
               "request_source_id": "UNIQUE-MESSAGE-1234",
               "request_source": "WEBAPP",
               "id": "1255",
               "trd_id": "null",
               "trade_ext_id": "",
               "status": "WARNING",
               "status_text": "Restricted Security - Pre-approval is required to trade this security"
```


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```
            }
        ],
            "success": true,
            "total_count": "1",
            "version": "1"
    }
```


Trade Cancellation

Successful cancellation of a trade.


```
{
  "items":  [
        {
                "ext_id": "25330",
                "ext_sys": "G1",
                "id": "59296",
                "src_id": "TA551BH",
                "src_sys": "TA",
                "status": "CANCELLED",
                "status_text": ""
          }
    ],
        "success": "true",
        "total_count": "1",
        "version": "1"
  }
```


Failure to cancel trade due to validation in Global One:


```
{
  "items":   [
        {
                "ext_id": "25332",
                "ext_sys": "G1",
                "id": "59298",
                "src_id": "TA551BK",
                "src_sys": "TA",
                "status": "NEW",
                "status_text": "Amendment or cancellation of Repo trades is prohibited"
          }
    ],
        "success": true,
        "total_count": 1,
        "version": 1
  }
```


Partial Return of a Trade

Failed to perform a partial return as the quantity was invalid, zero or negative.


```
{
  "errorMessage": "The Return Quantity value must be greater than zero.",
  "success": "false"
 }
```


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Full Return of a Trade

Failure to perform a full return as the request didn't contain identifiers to specific which trade to return.


```
{ "errorMessage": "Unable to find a trade. Need to specify id or src_sys&src_id or src_sys& src_
id_2 or ext_sys&ext_id.",
  "success": "false"
 }
```


Successful completion of a full return.


```
{
  "items":  [
              { "ext_id": "25320",
                "ext_sys": "G1",
                "id": "59286",
                "src_id": "TA550B5",
                "src_sys": "TA",
                "status": "PENDING",
                "status_text": ""
          }
    ],
        "success": "true",
        "total_count": "1",
        "version": "1"
  }
```


Fee Change of a Trade

Successful completion of a fee change:


```
{
  "items":  [
              { "fee": "80",
                "status_text": ""
          }
    ],
        "success": "true",
        "total_count": "1",
        "version": "1"
  }
```


Rate Change of a Trade

Successful completion of a rate change:


```
{
  "items":  [
              { "rebate": "0.4",
                "status_text": ""
          }
    ],
        "success": "true",
        "total_count": "1",
        "version": "1"
  }
```


Spread Change of a Trade

Successful completion of a spread change:

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```
{
  "items":  [
              { "spread": "3.0",
                "status_text": ""
          }
    ],
        "success": "true",
        "total_count": "1",
        "version": "1"
  }
```


Copy of a Trade

Successful completion of a trade copy:


```
{
  "items":  [
              { "worksheet": "Worksheet123",
                "status_text": ""
          }
    ],
        "success": "true",
        "total_count": "1",
        "version": "1"
  }
```


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